

Deep Learning-Based Anomaly Detection for Pixel-Level and Layered Image Attacks:

Research Proposal Report

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Abstract

Image-based attack vectors—maliciously perturbed QR codes and images composited with a hidden or transparent overlay layer—represent a growing and under-studied class of threats against automated visual verification systems. Unlike classical adversarial-example attacks that target a single deep learning classifier, these attacks exploit the gap between how a human perceives an image and how a downstream system decodes, classifies, or authenticates it. This report proposes a research program to design, implement, and evaluate a deep learning-based anomaly detection framework that (1) detects pixel-level perturbations engineered into QR codes to evade automated content-classification and phishing-detection pipelines, and (2) detects and localizes hidden or transparent overlay layers composited onto legitimate images, including estimation of how many layers are present and recovery of the payload contained in the concealed layer. The proposed approach combines high-frequency residual analysis, reconstruction-based anomaly detection, and invertible neural networks for layer separation, unified under a single anomaly-scoring pipeline. This report presents the problem statement, a structured literature review, research objectives, a proposed system architecture and methodology, candidate datasets and evaluation metrics, an experimental design, a research timeline, and a discussion of expected contributions and limitations.

1. Introduction

1.1 Background

Deep learning models are now embedded in security-critical image-processing pipelines: QR-code scanners that route users to payment pages or URLs, document- and ID-verification systems, and content-moderation filters that inspect uploaded images. These systems are typically trained and validated on the assumption that an input image is a single, unmodified visual artifact. In practice, an attacker can violate that assumption in at least two ways relevant to this research: (a) introducing small, often imperceptible pixel-level perturbations into an otherwise valid, scannable image (an adversarial-example-style attack), and (b) compositing a second, hidden layer—visually transparent or steganographically embedded—onto a legitimate-looking image, such that a human or a naive verifier perceives only the outer layer while a machine-readable payload is carried in the concealed layer.

Both attack styles share a common underlying mechanism: a deviation from the statistical properties of a 'clean' image manifold that is small enough to evade casual inspection but large enough to change how an automated system behaves. This shared mechanism motivates treating both problems under a common anomaly-detection research framework, while still allowing for attack-specific detection modules.

1.2 Motivation

QR codes are increasingly used as an attack surface for phishing ('quishing'), malware delivery, and payment fraud, precisely because they are opaque to casual human inspection and are usually trusted by scanning applications without visual scrutiny. Similarly, transparent-overlay and steganographic techniques have been used historically for covert communication and digital watermarking, but the

same mechanisms can be repurposed offensively: an attacker can hide a malicious payload, a spoofed identifier, or an adversarial pattern inside what appears to be an innocuous image layer, defeating systems that authenticate images by their visible content alone. As organizations increasingly deploy automated image-verification pipelines (KYC document checks, QR-based payment systems, automated content moderation), the practical consequence of undetected pixel-level and layered attacks grows correspondingly. This motivates dedicated research into detection methods rather than relying on defenses designed for unrelated threat models (e.g., network intrusion detection).

2. Problem Statement

Existing anomaly-detection and adversarial-defense research has largely developed along three separate tracks: (i) adversarial robustness for natural-image classifiers, (ii) steganalysis for covert-communication detection, and (iii) QR-code / barcode security, which remains comparatively under-researched at the pixel level. These tracks use different threat models, datasets, and evaluation protocols, which makes it difficult to build a single defensive pipeline for systems that must handle both structured machine-readable codes (QR codes) and natural or document images that may carry a hidden compositional layer.

This research addresses the following problem: given an input image that may be (a) a legitimately generated QR code, (b) a QR code that has been adversarially perturbed at the pixel level, (c) a legitimate single-layer image, or (d) an image composited from two or more layers (one of which may be transparent and carry a malicious or unauthorized payload), design a deep learning-based system that can (1) classify the input into one of these categories with high accuracy and low false-positive rate on legitimate inputs, (2) localize the specific pixels or regions responsible for the anomaly, and (3) where applicable, estimate the number of composited layers and reconstruct or characterize the content of the hidden layer.

3. Literature Review

3.1 Adversarial Attacks and Detection in Deep Learning Systems

Adversarial-example research has established that deep neural networks are vulnerable to small, carefully crafted input perturbations that cause misclassification while remaining imperceptible or near-imperceptible to humans. Detection-oriented defenses fall broadly into three families relevant to this research: (a) generative/reconstruction-based detectors that flag inputs with high reconstruction error or low likelihood under a model of the clean data distribution (e.g., PixelCNN-based purification, autoencoder-based reformers such as MagNet); (b) auxiliary-network detectors that attach binary classifiers to intermediate feature representations of the main model to distinguish clean from adversarial activations, an approach shown to achieve high detection accuracy with low false-positive rates on benchmark datasets; and (c) input-transformation/consistency detectors such as feature squeezing, which compare model outputs before and after a bit-depth or smoothing transformation and flag large disagreement as evidence of adversarial manipulation. A recurring finding across this literature is that adversarial detection performance is highly dependent on the

attacker's knowledge of the defense (white-box vs. black-box), which motivates evaluating any proposed detector under multiple attacker-capability assumptions rather than a single fixed attack.

3.2 QR Code and 'Quishing' Attack Detection

Research on QR-code-based phishing detection has so far focused primarily on the decoded payload—classifying the extracted URL using lexical and reputation-based features—rather than on the visual/pixel structure of the code itself. Recent work explicitly identifies robustness against adversarial perturbation of QR-code structural parameters (version, error-correction level, module styling) as an open research direction, noting that an attacker may manipulate these parameters to alter the code's visual pattern while preserving its decoded functionality. This leaves a gap that is directly relevant to the present research: there is little published work on detecting pixel-level tampering in the QR module grid itself, independent of the decoded payload, which is the first module proposed in Section 5.

3.3 Image Steganography and Steganalysis

Steganalysis research provides the closest existing methodological foundation for detecting a hidden or overlaid layer within an image. The field has moved from hand-crafted rich-model features combined with ensemble classifiers to end-to-end convolutional neural networks. A consistent architectural pattern across this literature is: (1) apply a high-pass filter to suppress natural image content and obtain a residual image dominated by embedding noise, and (2) train a CNN on this residual to classify the image as clean or stego. Architectures such as Ye-Net, DenseNet-based JPEG steganalyzers, and SRNet have progressively improved detection accuracy, with SRNet in particular used as a benchmark for evaluating detection of varying payload sizes (bytes of hidden data per pixel). Lightweight variants using MobileNetV2/EfficientNetV2S backbones with high-frequency preprocessing show that competitive accuracy is achievable at low computational cost, which is relevant to deployment in real-time QR-scanning or mobile verification contexts.

3.4 Layer Separation, Watermarking, and Composite-Image Forensics

Beyond binary stego/clean classification, a smaller body of work addresses recovering or characterizing the hidden content itself. Invertible neural networks (INNs) have recently been used to embed a learnable secondary signal within a host image such that the transformation is exactly invertible, allowing the hidden layer to be reconstructed at inspection time and compared against the visible layer to reveal tampering—demonstrated in the context of deepfake detection via a 'hidden face within a face.' Separately, fragile-watermarking approaches embed a signal at image-generation time specifically so that any later modification of the visible layer provably alters the watermark, providing a tamper-evidence mechanism rather than a purely post-hoc detector. These two approaches map onto two different deployment assumptions relevant to this research: INN-based layer recovery is applicable when the defender only has access to the (possibly already-composited) suspect image, whereas fragile watermarking is applicable when the defender controls the original image-generation pipeline (e.g., an organization's own QR-code generator).

3.5 Research Gap

Three gaps emerge from this review. First, pixel-level (as opposed to payload-level) detection of adversarially perturbed QR codes is largely unexplored. Second, steganalysis research is overwhelmingly evaluated on natural photographic images (e.g., BOSSBase, ALASKA2) rather than on structured machine-readable codes or mixed document/QR content, so its transferability to this threat model is untested. Third, existing layer-recovery methods (INN-based) have been demonstrated for specific applications (deepfake face-tampering) but not generalized into a layer-count estimation and payload-characterization pipeline applicable to arbitrary transparent-overlay attacks. This research is positioned to address these three gaps.

4. Research Objectives

The overall aim of this research is to design and evaluate a deep learning-based anomaly detection framework for pixel-level and layered image attacks. The specific objectives are:

1. To design and train a pixel/module-level anomaly detector for QR codes capable of distinguishing legitimately generated codes from adversarially perturbed variants, independent of whether the decoded payload appears benign.
2. To design and evaluate a residual-based deep learning classifier for detecting the presence of a hidden or transparent overlay layer composited onto a legitimate image.
3. To develop a layer-depth estimation method that predicts the number of composited layers present in an input image and localizes the spatial regions attributable to each layer.
4. To investigate invertible neural network architectures for reconstructing or characterizing the content of a detected hidden layer, to support forensic follow-up.
5. To integrate objectives 1–4 into a unified anomaly-scoring pipeline and evaluate it end-to-end under both clean and adversarial/attacker-aware conditions, including a false-positive-rate analysis on legitimate, unmodified inputs.

5. Proposed Methodology

5.1 System Overview

The proposed system consists of four coordinated modules operating on an input image: (M1) a QR-structure validator and pixel-anomaly detector, (M2) a residual-based steganographic/overlay detector, (M3) a layer-depth estimator, and (M4) an invertible-network-based layer separator. Module routing is determined by a lightweight input-type classifier that first distinguishes QR-code inputs from general image inputs, since QR codes have a known geometric structure that can be exploited directly, while general images require the residual/statistical approach used in steganalysis.

5.2 Module 1 — QR Pixel-Level Anomaly Detection

This module exploits the fact that a valid QR code has a fully specified geometric and error-correction structure (finder patterns, alignment patterns, timing patterns, quiet zone, and a bounded module grid). The proposed pipeline is:

- Grid rectification: detect the QR bounding box and finder patterns, then apply a perspective transform to normalize the code into a canonical module grid.
- Structural consistency check: compare each module's measured value against the value expected from the decoded error-correction codeword; flag modules that fall outside expected black/white thresholds without being explainable by print/scan noise.
- Learned anomaly scoring: train a convolutional autoencoder exclusively on a large corpus of legitimately generated QR codes (varying content, size, and error-correction level); at inference time, per-module reconstruction error above a calibrated threshold flags a pixel-level anomaly.
- Cross-path verification: independently decode the payload via a standard decoder and via a CNN classifier trained on raw pixel grids; a disagreement between the two paths is treated as a strong anomaly signal, following the feature-squeezing principle of comparing outputs across two independent processing paths.

Adversarial QR samples for training and evaluation will be generated synthetically by applying gradient-based perturbation methods (e.g., FGSM/PGD-style pixel perturbations constrained to preserve decodability) to a base set of legitimately generated codes, producing paired clean/adversarial examples with known ground-truth perturbation masks.

5.3 Module 2 — Overlay/Steganographic Layer Detection

This module follows the established steganalysis pattern but is adapted to the transparent-overlay threat model rather than only LSB-style embedding:

- High-pass residual extraction using a fixed or learned high-pass filter bank (SRM-style filters) to suppress semantic image content and expose composition artifacts.
- A CNN classifier (SRNet-style, with a lightweight MobileNet/EfficientNet variant for deployment-constrained settings) trained on the residual to output a binary clean/composited label and a confidence score.
- Spatial localization via class-activation mapping or a segmentation head, to highlight which regions of the image are attributable to the overlay layer rather than the base image.

5.4 Module 3 — Layer-Depth Estimation

Layer depth, in this research, refers to the number of distinct composited layers detected in an image and their relative stacking order — not 3D scene depth. This is framed as a multi-class / count-regression problem: given an input flagged as composited by Module 2, a secondary network estimates the number of layers (e.g., 1 = clean, 2 = single overlay, 3+ = multiple stacked overlays) and produces a per-pixel layer-assignment map. Training data will be synthetically generated by compositing controlled numbers of transparent layers (varying opacity, blend mode, and payload type) onto base images, giving exact ground-truth layer counts and masks.

5.5 Module 4 — Invertible-Network Layer Separation

For images flagged as composited, an invertible neural network will be trained to approximate the inverse of the compositing function, recovering an estimate of the hidden layer's content for forensic inspection. This follows the recent 'hidden-signal-within-a-host' paradigm used in deepfake tamper detection, adapted here to general overlay payloads rather than a fixed secret image. The

bijection property of the INN allows a single trained network to serve both the forward (embedding, for generating synthetic training data) and inverse (recovery, for detection) directions.

5.6 Unified Anomaly-Scoring Pipeline

The four modules will be combined into a single pipeline: input-type routing, followed by parallel anomaly scoring, followed by a fusion layer (a lightweight gradient-boosted or shallow neural fusion model) that combines module-level scores into a final risk classification (clean / adversarially perturbed / overlay-composited / both) with an associated confidence and localization map. This fusion stage allows the system to be evaluated both per-module and end-to-end.

6. Datasets

The following datasets and data-generation strategies are proposed:

Purpose	Source / Generation Strategy
Clean QR corpus	Synthetically generated QR codes spanning versions 1–40, all four error-correction levels, and varied embedded payload content and physical print/scan simulation (blur, rotation, lighting).
Adversarial QR corpus	FGSM/PGD-style pixel perturbations applied to the clean corpus under a decodability constraint; parameter-level manipulations (version/EC-level/styling) as identified in current quishing literature.
Steganalysis benchmark	BOSSBase, BOWS2, and ALASKA2 for baseline steganalysis model validation against established literature results.
Overlay/composite corpus	Synthetically composited images (transparent-layer blending at varying opacity, JPEG re-compression, and common image-processing operations) built from a base natural-image dataset (e.g., a public document/ID or general-purpose image dataset appropriate to the target deployment domain).
Real-world validation set	A held-out set of QR codes and document images collected from public/legitimate sources, used only for false-positive-rate evaluation, not training.

Table 1. Proposed datasets and their role in the experimental design.

7. Evaluation Metrics

Detection performance will be reported using standard classification metrics — accuracy, precision, recall, F1-score, and area under the ROC curve — computed separately for each module and for the fused pipeline. In addition, the following domain-specific metrics are proposed:

- False-positive rate on the real-world legitimate validation set, reported separately from benchmark-dataset performance, since deployment usability depends primarily on this figure.
- Localization accuracy (Intersection-over-Union) between predicted anomaly/overlay masks and ground-truth perturbation or compositing masks.
- Layer-count estimation accuracy (exact match and mean absolute error) for Module 3.

- Reconstruction fidelity (e.g., PSNR/SSIM between recovered and ground-truth hidden-layer content) for Module 4.
- Robustness under attacker-aware conditions: detection performance when the attacker has full knowledge of the detector (white-box adaptive attack) versus no knowledge (black-box), consistent with adversarial-detection evaluation practice.

8. Experimental Design

Experiments are organized into four phases, each producing a deliverable that feeds the next:

6. Baseline reproduction: reproduce published steganalysis results (SRNet or equivalent) on BOSSBase/ALASKA2 to validate the experimental pipeline before adapting it to the QR and overlay-specific datasets.
7. Module-level training and ablation: train Modules 1–4 independently, with ablation studies isolating the contribution of the high-pass residual step, the cross-path verification step, and the choice of backbone architecture (lightweight vs. high-capacity).
8. Fusion and end-to-end evaluation: train the fusion layer on module outputs and evaluate the complete pipeline on held-out synthetic and real-world data.
9. Adversarial robustness evaluation: re-attack the trained pipeline using adaptive, detector-aware perturbations to measure worst-case performance degradation, and report results under both white-box and black-box attacker assumptions.

9. Expected Outcomes and Contributions

This research is expected to contribute: (1) a pixel-level anomaly-detection method for QR codes that operates independently of decoded-payload analysis, addressing an explicitly identified gap in current quishing-detection literature; (2) an empirical evaluation of established steganalysis architectures on a transparent-overlay threat model distinct from their original LSB-embedding evaluation setting; (3) a layer-depth estimation method extending binary stego/clean detection to a count-and-localize formulation; (4) an applied evaluation of invertible-network-based layer recovery outside its original deepfake-detection context; and (5) an open, documented synthetic dataset generation pipeline for adversarial-QR and transparent-overlay training data, which may be of independent value to the research community given the current scarcity of public datasets for this specific threat model.

10. Research Timeline

Phase	Duration	Milestone
Literature review & problem formalization	Weeks 1–4	Finalized related-work chapter and formal threat model
Dataset construction (clean + adversarial QR, composite images)	Weeks 5–9	Synthetic dataset pipeline and documentation
Module 1 & 2 development	Weeks 10–15	Trained QR anomaly detector and overlay detector with baseline metrics

Phase	Duration	Milestone
Module 3 & 4 development	Weeks 16–21	Layer-depth estimator and INN-based recovery prototype
Fusion pipeline & end-to-end evaluation	Weeks 22–25	Integrated pipeline with full metric suite
Adversarial robustness evaluation	Weeks 26–28	White-box/black-box robustness results
Thesis writing and defense preparation	Weeks 29–32	Final thesis draft and defense

Table 2. Proposed 32-week research timeline.

11. Limitations and Ethical Considerations

This research is defensive in nature: all adversarial and overlay-attack samples will be generated solely for training and evaluating detection models, not for deployment against third-party systems. Synthetic attack generation will be restricted to controlled datasets, and any real-world data collection will exclude personally identifiable or sensitive document content unless explicit consent and institutional ethical approval are obtained. A methodological limitation is that detector performance against adaptive, detector-aware attackers is inherently bounded by the attacker's assumed knowledge of the system; this research addresses that limitation by explicitly evaluating under both white-box and black-box conditions rather than reporting only best-case detection rates. Finally, results obtained on synthetic composite-image datasets may not fully transfer to attacker techniques not represented in the training distribution; the real-world validation set (Section 6) is included specifically to surface this generalization gap rather than to eliminate it.

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Note: Full bibliographic entries (venue, volume, page numbers) should be completed and formatted according to the citation style required by the student's institution (e.g., IEEE, APA) prior to submission. Additional primary sources on PixelDefend, MagNet, and feature squeezing should be added directly from the original publications during the literature-review expansion phase.